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Classifying Human Values in Social Media Posts:

A GPT-Assisted Multi-Stage Pipeline for Non-English Textual Data

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Motivation

- Studying values in social media content helps to illuminate societal change
- Values are abstract and subjective – difficult to detect in online communication
- Most existing studies focus on user-level profiling, and rely on English-language or curated datasets

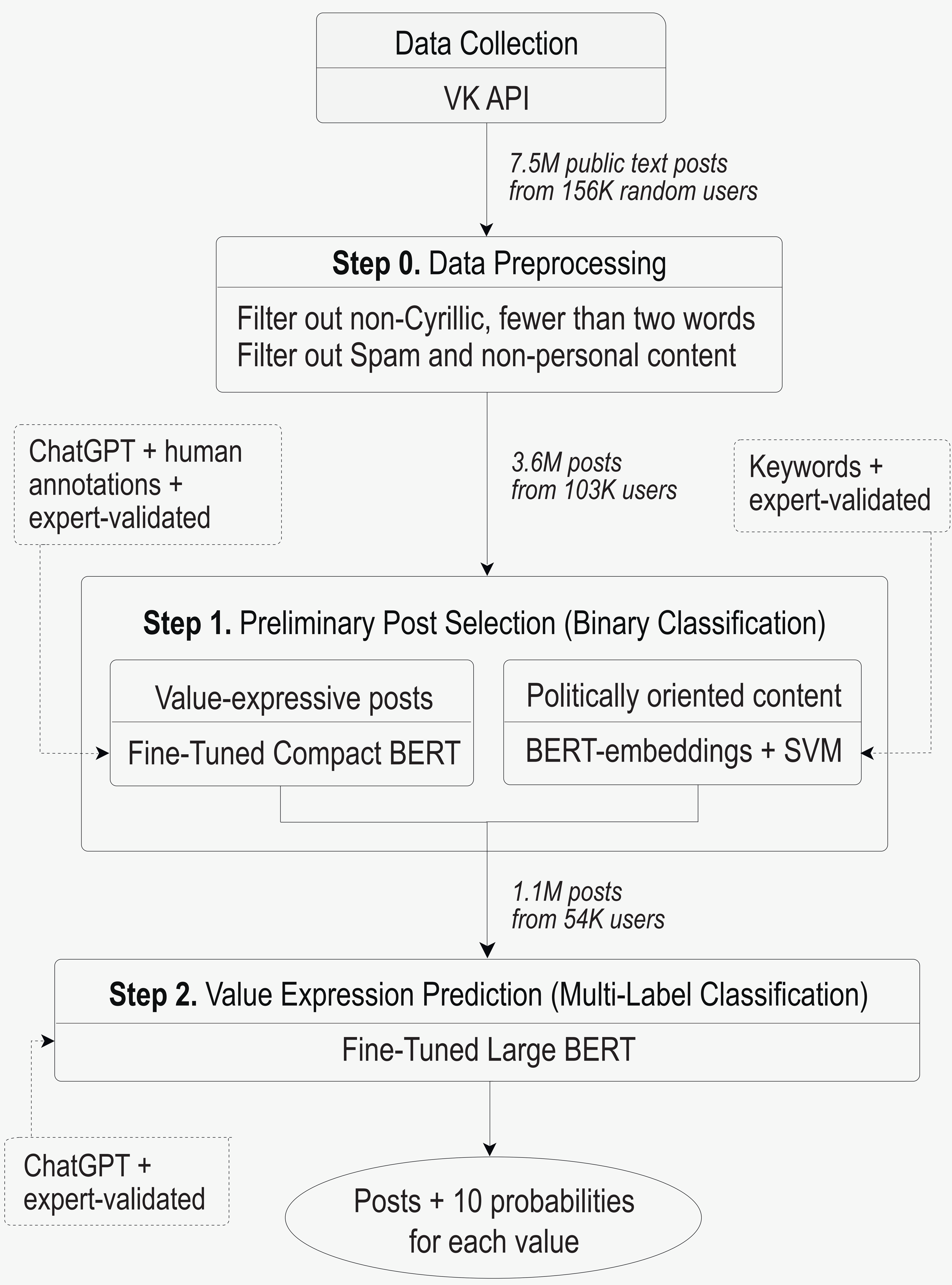
▶ ***Main goal:*** *Develop a pipeline for classifying human values in large-scale data to capture "value climate" of social media.*

DATA

- Platform: VKontakte (largest social media platform in Russia)
- 1M random user IDs ⟶ 16% IDs have public timelines ⟶ 7.5M text posts (2007–2025)
- Representative sample of users' self-expression
- Dataset fully anonymized (hashed IDs, no personal info)

Methodological Framework

- Schwartz's (1992) theory of 10 basic human values
- Multi-label classification (a post may express several values)
- Focus on value expression in texts, not on users' personal values
- BERT-like models, human and AI annotators



Results

▶ Methodological Findings

Annotation Validation

- Interrater agreement: not high ~ highlights the abstract nature of value expression (Step 1: $\alpha_{crowd} \approx 0.4$, $\alpha_{experts} \approx 0.6$; Step 2: $\alpha_{experts} \approx 0.3$). GPT annotations were more consistent (Step 1: $\alpha_{gpt} \approx 0.95$; Step 2: $\alpha_{gpt} \approx 0.6$)
- Internal consistency of GPT judgments mirrored expert agreement levels
- Validation: *ICC = 0.7* (mean expert and GPT scores); *Accuracy ≈ 0.8, F1 ≈ 0.6* (expert and GPT majority vote)
- To address subjectivity, GPT annotations were aggregated into soft labels (1.0 / 0.6 / 0.0), depending on the level of agreement

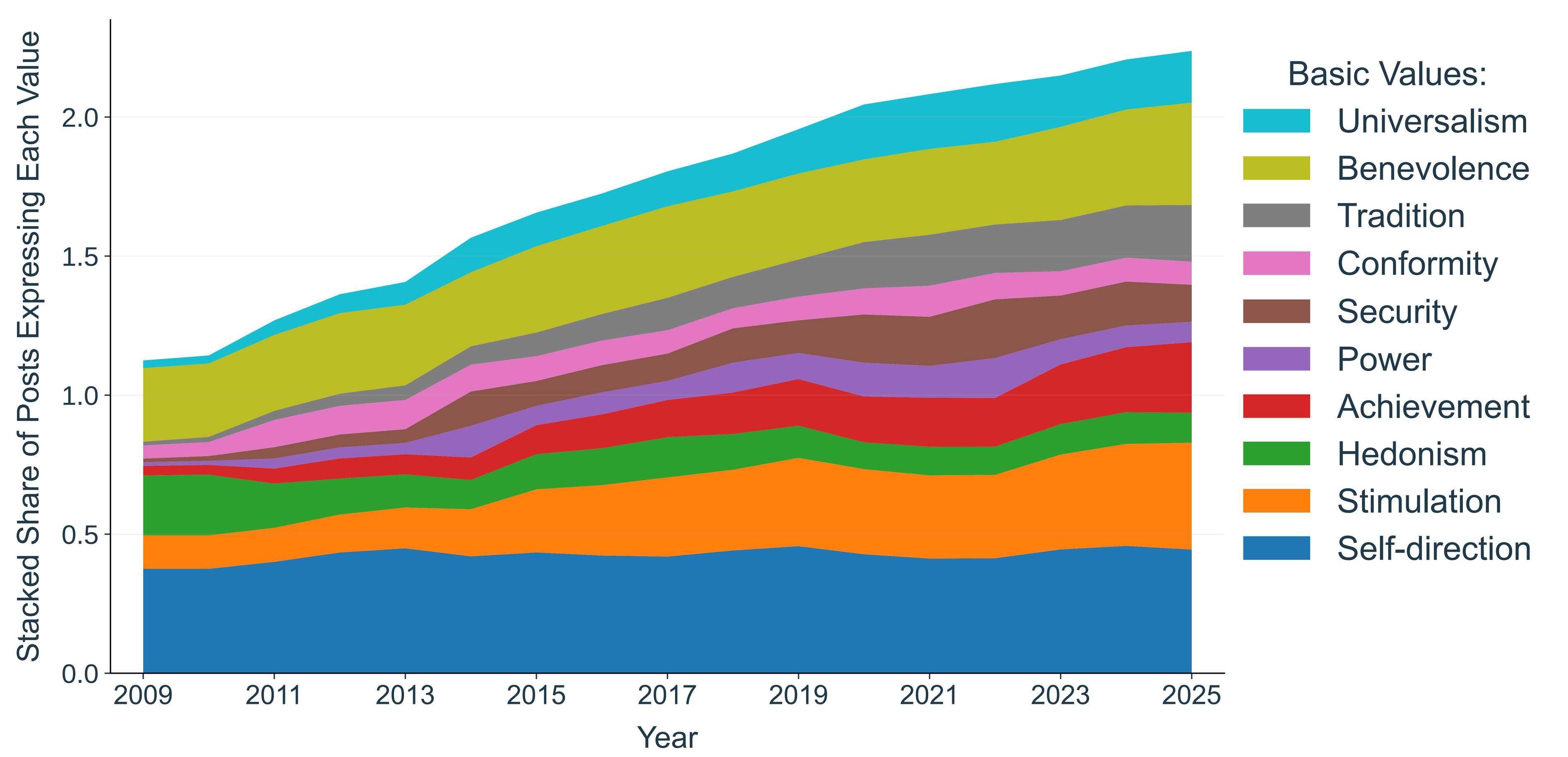
Model Performance across Steps

Step	Best Model	F1 / F1-macro
Step 0. Spam Filtering	TF-IDF + Logistic Regression	0.96 / 0.97
Step 1. Binary classification:		
Value Detection	RuBERT-tiny + SVM Classifier	0.77 / 0.83
Political Relevance	DistilRuBERT + SVM Classifier	0.82 / 0.91
Step 2. Multi-Label Classification	XLM-RoBERTa-large (soft labels)	0.71 / 0.83

▶ Empirical Findings

- ~30% of posts are value-expressive or politically oriented
- Overall trend: value expressiveness increased over time
- Long-term stability: Self-Direction and Benevolence remain most prevalent
- Temporal shifts: Increases in Stimulation, Achievement, Security, Tradition, and Universalism
- Correlation analysis:
  - Replicates Schwartz's structure (Security–Tradition–Conformity, Self-Direct.–Stimulation)
  - Some values take locally specific meanings, e.g., Universalism is often expressed in connection with Tradition and Security rather than as concern for all people

Dynamics of Value Expression on Social Media, 2009–2025



Policy Implications: Scalable Method Beyond Surveys

- Monitoring of basic values when mass surveys become unavailable or biased
- Providing a scalable way to study closed or hard-to-reach populations
- Detection of long-term shifts in society while not requiring expensive methods
- Informing strategic decisions in geopolitics and global business (as global agendas should adapt to local narratives)
- The pipeline is extendable to other languages, platforms, and populations.